Course 6: Multi-Cloud Architecture: AWS & Azure High Availability

Multi-Cloud High Availability & Disaster Recovery Guide

Official Study Guide • Instructor: Dr. Vikram Sen (Principal Cloud Solutions Architect) • EduPulse Academy

1. Multi-Region Active-Active Routing & Global Load Balancing

Active-Active deployments require DNS latency-based routing (AWS Route 53 / Azure Traffic Manager) combined with health checks to automatically failover traffic within seconds.

Stateless application tiers backed by auto-scaling container groups (AWS ECS / Azure Kubernetes Service) allow seamless geographic load distribution.

```
resource "aws_route53_record" "global_app" {
  zone_id = var.dns_zone_id
  name    = "api.edupulse.com"
  type    = "A"
  latency_routing_policy {
    region = "us-east-1"
  }
  alias {
    name                   = aws_lb.east_alb.dns_name
    zone_id                = aws_lb.east_alb.zone_id
    evaluate_target_health = true
  }
}
```

2. Distributed Database Synchronization & CAP Theorem

Under network partitions (CAP theorem), systems must trade off immediate consistency for high availability. DynamoDB Global Tables and Azure Cosmos DB provide multi-region multi-master writes with tunable consistency levels (Session, Bounded Staleness, Eventual).

Idempotency keys on REST APIs prevent double-charging or duplicate record writes during transient network retries across cloud boundaries.

3. Zero-Trust Security & Multi-Cloud Identity Federation

Enforce mutual TLS (mTLS) across microservices and manage workload identities via OIDC federation (AWS IAM Roles for Service Accounts / Azure Workload Identity) rather than hardcoded long-lived credentials.

All sensitive configuration strings must be encrypted in transit and at rest using cloud HSM key vaults (AWS KMS / Azure Key Vault) with strict audit access logging.

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